

Kilopower 10 kWe Stirling Nuclear Reactor Operating Limits

1. Reactor Core Architecture

The nuclear power assembly utilizes an enriched uranium-235 molybdenum alloy core casting enclosed in a beryllium oxide reflector ring, operating at nominal core temperature of 800 degrees Celsius.

2. Stirling Power Conversion

Sodium heat pipes transfer thermal energy to dual balanced Stirling engines, producing 10 kilowatts continuous electrical power with design operational lifespan of 10 Martian years.

3. Radiation Shielding Zone

A lithium hydride and depleted tungsten shadow shield attenuates gamma and neutron radiation to under 5 rem/year at the habitat perimeter boundary 100 meters away.